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MATHEMATICS

0580/22

Paper 2 (Extended)

May/June 2020

1 hour 30 minutes

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

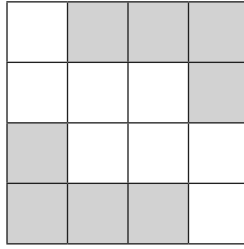
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should use a calculator where appropriate.
- You may use tracing paper.
- You must show all necessary working clearly.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages. Blank pages are indicated.

1



Write down the order of rotational symmetry of the diagram.

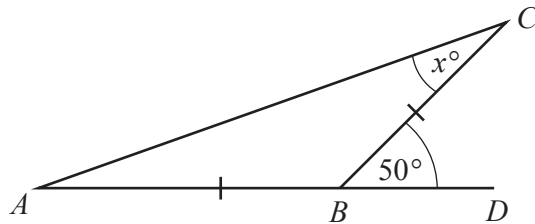
..... [1]

- 2 At noon the temperature in Maseru was 21°C .
At midnight the temperature had fallen by 26°C .

Work out the temperature at midnight.

..... $^{\circ}\text{C}$ [1]

3



NOT TO
SCALE

$AB = BC$ and ABD is a straight line.

Find the value of x .

$x =$ [2]

- 4 Write down

(a) a square number greater than 10,

..... [1]

(b) an irrational number.

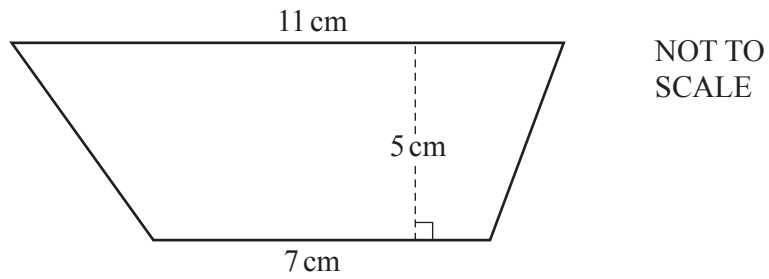
..... [1]

5 $y = mx + c$

Find the value of y when $m = -3$, $x = -2$ and $c = -8$.

$y =$ [2]

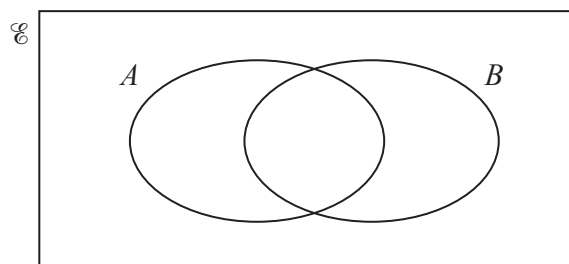
6



Calculate the area of the trapezium.

..... cm^2 [2]

7



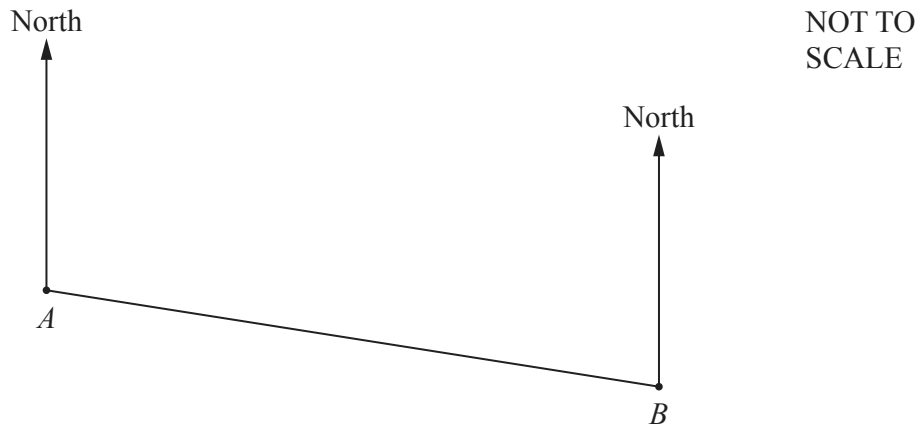
On the Venn diagram, shade the region $A \cap B$.

[1]

8 Write 2^{-4} as a decimal.

..... [1]

9



The bearing of B from A is 105° .

Find the bearing of A from B .

..... [2]

10 Simplify.

$$\frac{p}{2q} \times \frac{4pq}{t}$$

..... [2]

11 **Without using a calculator**, work out $1\frac{3}{4} - \frac{11}{12}$.

You must show all your working and give your answer as a fraction in its simplest form.

..... [3]

- 12** Roberto buys a toy for \$5.00 .
He then sells it for \$4.60 .

Calculate his percentage loss.

..... % [2]

- 13** Simplify $8t^8 \div 4t^4$.

..... [2]

- 14** Solve the equation.

$$\frac{1-x}{3} = 5$$

$x =$ [2]

- 15** Ella's height is 175 cm, correct to the nearest 5 cm.

Write down the upper bound of Ella's height.

..... cm [1]

- 16** Calculate $(3 \times 10^{-3})^3$.
Give your answer in standard form.

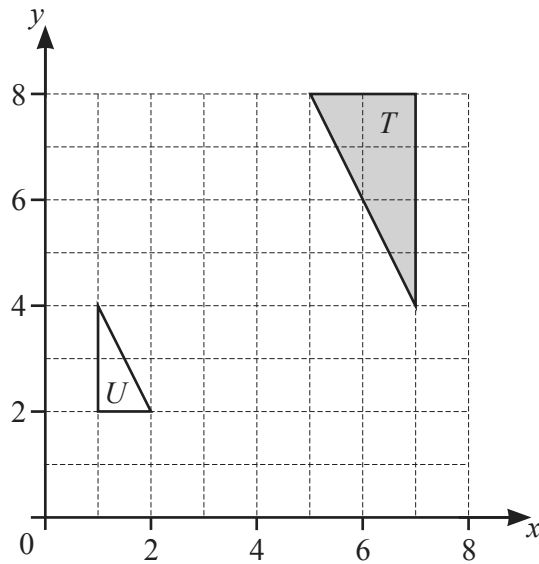
..... [1]

- 17 A train of length 105 m takes 11 seconds to pass completely through a station of length 225 m.

Calculate the speed of the train in km/h.

..... km/h [3]

18



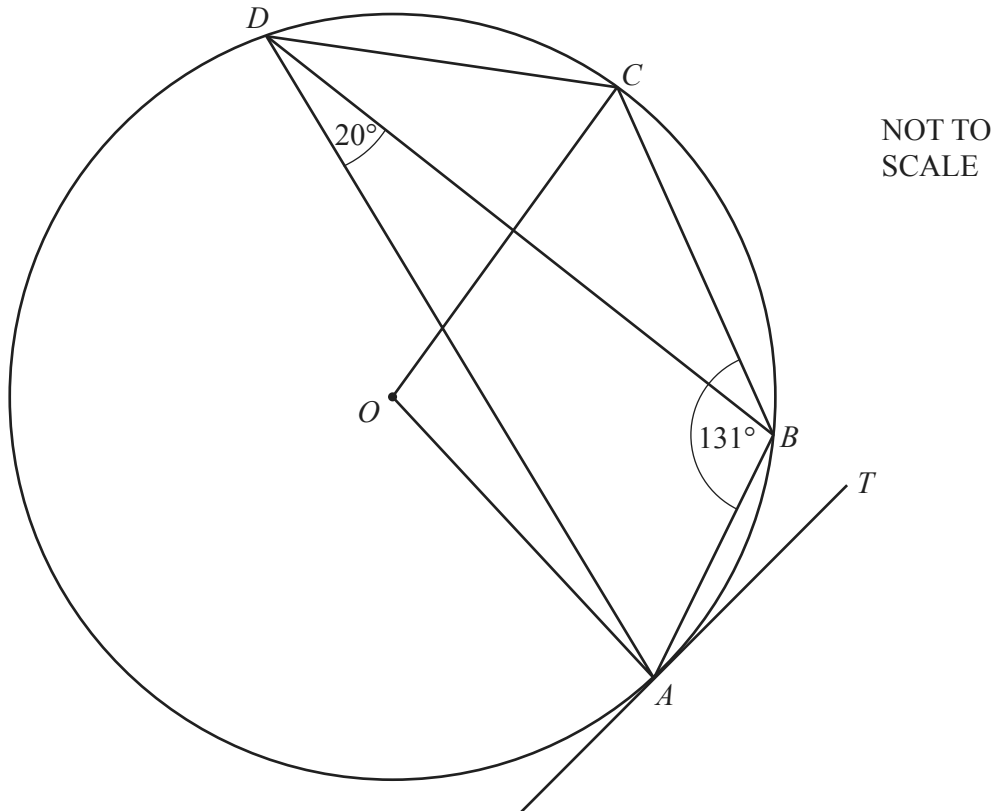
Describe fully the **single** transformation that maps triangle T onto triangle U .

.....
 [3]

- 19 Make y the subject of the formula.

$$h^2 = x^2 + 2y^2$$

$y =$ [3]



A, B, C and D lie on the circle, centre O .
 TA is a tangent to the circle at A .
 Angle $ABC = 131^\circ$ and angle $ADB = 20^\circ$.

Find

(a) angle ADC ,

Angle $ADC = \dots\dots\dots$ [1]

(b) angle AOC ,

Angle $AOC = \dots\dots\dots$ [1]

(c) angle BAT ,

Angle $BAT = \dots\dots\dots$ [1]

(d) angle OAB .

Angle $OAB = \dots\dots\dots$ [1]

21 Simplify.

(a) $(5x^4)^3$

..... [2]

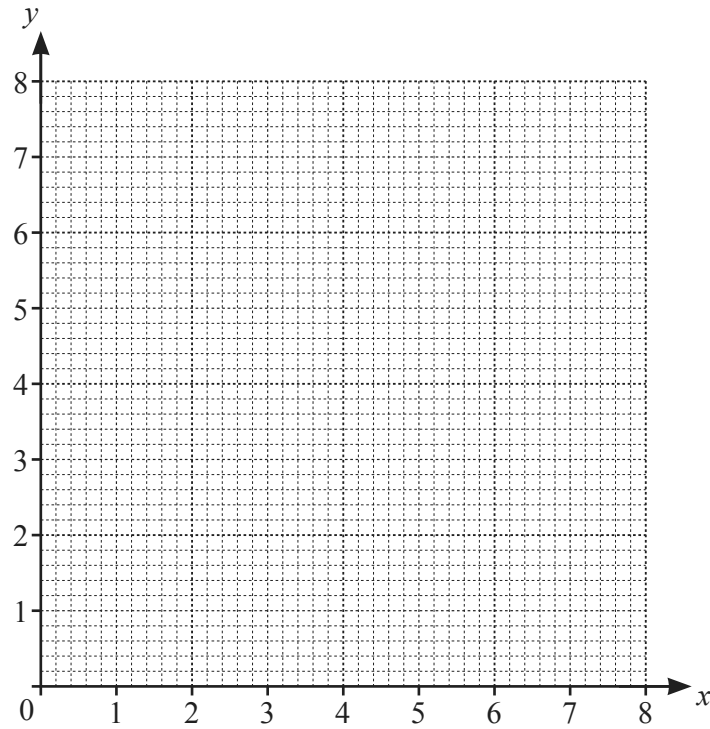
(b) $(256x^{256})^{\frac{3}{8}}$

..... [2]

22 p is directly proportional to $(q + 2)^2$.
When $q = 1$, $p = 1$.

Find p when $q = 10$.

$p =$ [3]



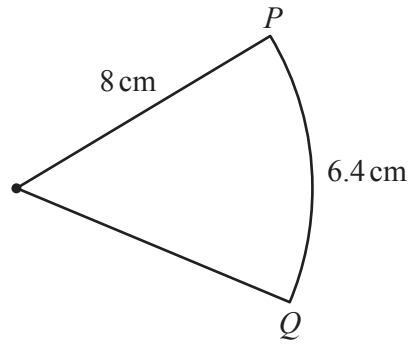
- (a) By drawing suitable lines and shading unwanted regions, find the region, R , where

$$x \geq 2, \quad y \geq x \quad \text{and} \quad 2x + y \leq 8. \quad [5]$$

- (b) Find the largest value of $x + y$ in the region R .

..... [1]

24

NOT TO
SCALE

The diagram shows a sector of a circle of radius 8 cm.
The length of the arc PQ is 6.4 cm.

Find the area of the sector.

..... cm^2 [4]

25 Simplify.

$$\frac{2x^2 + x - 15}{ax + 3a - 2bx - 6b}$$

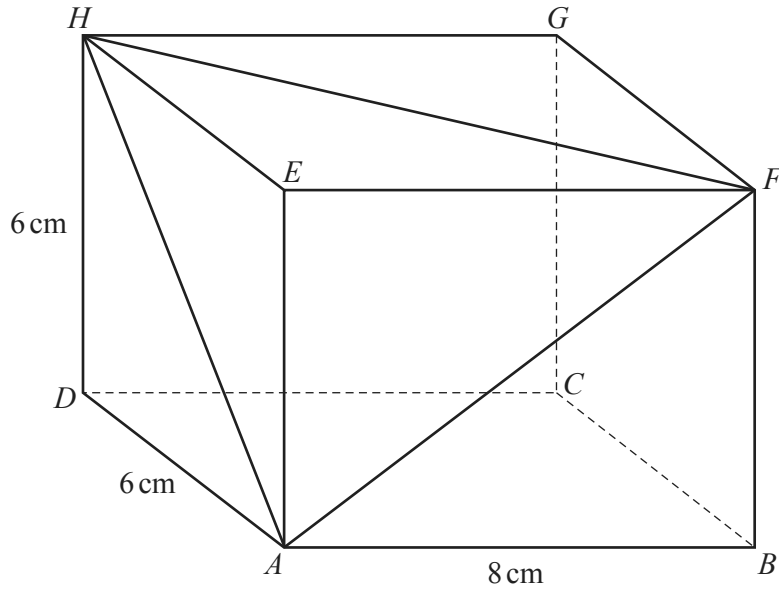
..... [5]

26 $\sqrt[3]{y^2} = \sqrt[n]{x}$ and $y = \sqrt[n]{x}$.

Find the value of n .

$n =$ [2]

Question 27 is printed on the next page.



NOT TO
SCALE

The diagram shows a cuboid.

$AB = 8\text{ cm}$, $AD = 6\text{ cm}$ and $DH = 6\text{ cm}$.

Calculate angle HAF .

Angle $HAF = \dots\dots\dots$ [6]

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